

LW HW4 Coding

September 28, 2025

```
[1]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
import statsmodels.stats.anova as anova
from statsmodels.stats.outliers_influence import summary_table
from scipy import stats
import matplotlib.pyplot as plt
```

```
[2]: print("="*78)
print("CODING QUESTION 4: KelleyBlueBookData Dataset Analysis")
print("="*78)
```

```
=====
CODING QUESTION 4: KelleyBlueBookData Dataset Analysis
=====
```

```
[3]: data = pd.read_csv('/Users/lokeshmuvva/Documents/F25_1/regression_f25/data/
↳KelleyBlueBookData.csv')
print(data.head())
```

	Price	Mileage	Make	Model	Trim	Type	Cylinder	Liter	\
0	17314.10313	8221	Buick	Century	Sedan 4D	Sedan	6	3.1	
1	17542.03608	9135	Buick	Century	Sedan 4D	Sedan	6	3.1	
2	16218.84786	13196	Buick	Century	Sedan 4D	Sedan	6	3.1	
3	16336.91314	16342	Buick	Century	Sedan 4D	Sedan	6	3.1	
4	16339.17032	19832	Buick	Century	Sedan 4D	Sedan	6	3.1	

	Doors	Cruise	Sound	Leather
0	4	1	1	1
1	4	1	1	0
2	4	1	1	0
3	4	1	0	0
4	4	1	0	1

```
[4]: X = data[['Mileage', 'Liter', 'Cylinder']]
y = data['Price']

X = sm.add_constant(X)
```

```
[5]: # Fit the model
model = smf.ols('Price ~ Mileage + Liter + Cylinder',data=data).fit()
```

```
[6]: seq = sm.stats.anova_lm(model, typ=1)
seq
```

	df	sum_sq	mean_sq	F	PR(>F)
Mileage	1.0	1.605590e+09	1.605590e+09	24.890171	7.448191e-07
Liter	1.0	2.421824e+10	2.421824e+10	375.435819	7.134623e-69
Cylinder	1.0	1.031948e+09	1.031948e+09	15.997457	6.930502e-05
Residual	800.0	5.160560e+10	6.450701e+07	NaN	NaN

```
[7]: print("="*50)
print("PART (a): Model Interpretation")
print("="*50)

# Extract F and p-value for Cylinder
f_val = seq.loc["Cylinder", "F"]
p_val = seq.loc["Cylinder", "PR(>F)"]

print(f"\nF value for Cylinder: {f_val:.4f}, p-value: {p_val:.4g}")

print("\nInterpretation:")
print("H: _Cylinder = 0 (Cylinder adds no predictive value beyond Mileage and
↳Liter)")
print("H: _Cylinder  0 (Cylinder adds significant predictive value beyond
↳Mileage and Liter)")
if p_val < 0.05:
    print(f"P-value = {p_val:.6f} < 0.05, so we reject H")
    print("Conclusion: Adding Cylinder significantly improves the model that
↳already contains Mileage and Liter")
else:
    print(f"P-value = {p_val:.6f}  0.05, so we fail to reject H")
    print("Conclusion: Adding Cylinder does not significantly improve the model
↳that already contains Mileage and Liter")
```

```
=====
PART (a): Model Interpretation
=====
```

F value for Cylinder: 15.9975, p-value: 6.931e-05

Interpretation:

H: _Cylinder = 0 (Cylinder adds no predictive value beyond Mileage and Liter)

H: _Cylinder 0 (Cylinder adds significant predictive value beyond Mileage and Liter)

P-value = 0.000069 < 0.05, so we reject H

Conclusion: Adding Cylinder significantly improves the model that already

contains Mileage and Liter

```
[8]: print("="*50)
print("PART (b): Manual ANOVA Test")
print("="*50)

null_model = smf.ols('Price ~ Mileage + Liter',data=data).fit()
full_model = smf.ols('Price ~ Mileage + Liter + Cylinder',data=data).fit()

seq_null = sm.stats.anova_lm(null_model, typ=1)
seq_full = sm.stats.anova_lm(full_model, typ=1)

print("Null Model")
print(seq_null)
print("\nFull Model")
print(seq_full)

SSE_H0 = seq_null.loc['Residual', 'sum_sq']
df_H0 = seq_null.loc['Residual', 'df']

SSE_H1 = seq_full.loc['Residual', 'sum_sq']
df_H1 = seq_full.loc['Residual', 'df']

print(f"\nSSE_H0 = {SSE_H0}")
print(f"df_H0 = {df_H0}")
print(f"SSE_H1 = {SSE_H1}")
print(f"df_H1 = {df_H1}")

numerator = (SSE_H0 - SSE_H1) / (df_H0 - df_H1)
denominator = (SSE_H1/df_H1)

print(f"\nNumerator = ({SSE_H0} - {SSE_H1})/({df_H0} - {df_H1})")
print(f"Denominator = ({SSE_H1}/{df_H1})")

manual_F_stat = numerator/denominator

print(f"\nThe manually calculated F statistic is {manual_F_stat:.4f}")
print(f"The F-value from the original table is {f_val:.4f}")
if np.isclose(manual_F_stat, f_val):
    print("\nThis matches the F-statistic calculated from part (a)")
else:
    print("\nThis does not match the F-statistic calculated from part (a)")
```

=====
PART (b): Manual ANOVA Test
=====

Null Model	df	sum_sq	mean_sq	F	PR(>F)
------------	----	--------	---------	---	--------

Mileage	1.0	1.605590e+09	1.605590e+09	24.432707	9.374438e-07
Liter	1.0	2.421824e+10	2.421824e+10	368.535574	7.328735e-68
Residual	801.0	5.263755e+10	6.571480e+07	NaN	NaN

Full Model

	df	sum_sq	mean_sq	F	PR(>F)
Mileage	1.0	1.605590e+09	1.605590e+09	24.890171	7.448191e-07
Liter	1.0	2.421824e+10	2.421824e+10	375.435819	7.134623e-69
Cylinder	1.0	1.031948e+09	1.031948e+09	15.997457	6.930502e-05
Residual	800.0	5.160560e+10	6.450701e+07	NaN	NaN

SSE_H0 = 52637552166.235855

df_H0 = 801.0

SSE_H1 = 51605604120.22623

df_H1 = 800.0

Numerator = (52637552166.235855 - 51605604120.22623)/(801.0 - 800.0)

Denominator = (51605604120.22623/800.0)

The manually calculated F statistic is 15.9975

The F-value from the original table is 15.9975

This matches the F-statistic calculated from part (a)

```
[9]: print("="*50)
print("PART (c): Partial (Type II) Sum of Squares")
print("="*50)

# Type II (Partial) ANOVA
anova_type2 = anova.anova_lm(model, typ=2)
print("Type II (Partial) ANOVA:")
print(anova_type2)

cylinder_f_type2 = anova_type2.loc['Cylinder', 'F']
cylinder_p_type2 = anova_type2.loc['Cylinder', 'PR(>F)']

print(f"\nF-value for Cylinder from partial ANOVA (typ=2): {cylinder_f_type2:.6f}")
print(f"P-value for Cylinder from partial ANOVA (typ=2): {cylinder_p_type2:.6f}")

print(f"\nF-value for Cylinder from part (a): {f_val:.4f}, p-value: {p_val:.4g}")

if np.isclose(cylinder_f_type2, f_val) and np.isclose(cylinder_p_type2, p_val):
    print("\nThis matches the F-test calculated from part (a)")
else:
```

```
print("\nThis does not match the F-test calculated from part (a)")
```

```
=====
PART (c): Partial (Type II) Sum of Squares
=====
```

```
Type II (Partial) ANOVA:
```

	sum_sq	df	F	PR(>F)
Mileage	1.283997e+09	1.0	19.904763	0.000009
Liter	1.929760e+08	1.0	2.991552	0.084086
Cylinder	1.031948e+09	1.0	15.997457	0.000069
Residual	5.160560e+10	800.0	NaN	NaN

```
F-value for Cylinder from partial ANOVA (typ=2): 15.997457
```

```
P-value for Cylinder from partial ANOVA (typ=2): 0.000069
```

```
F-value for Cylinder from part (a): 15.9975, p-value: 6.931e-05
```

```
This matches the F-test calculated from part (a)
```

The F-stat for Cylinder is the same in both cases because of the order in which both types are arranged when deriving these values. In $\text{typ} = 1$, Cylinder is last in the order, so the full model adds Cylinder after already including Mileage and Liter (in that order). In $\text{typ} = 2$, the full model adds Cylinder after controlling for the other predictors. The F-statistic is the same because Cylinder is tested last in both approaches—sequentially after Mileage and Liter in Type I, and while controlling for both predictors in Type II.

```
[10]: print("="*50)
print("PART (d): Partial (Type II) Sum of Squares")
print("="*50)

print("\nInterpretation of 'Mileage' in Type II ANOVA:")
print("- Reduced model: Price ~ Liter + Cylinder")
print("- Full model: Price ~ Mileage + Liter + Cylinder")
print("- Partial SS tests: Is 'Mileage' needed given that 'Liter' and
    ↳ 'Cylinder' are in the model?")
print("- H : _Mileage = 0 (given Liter and Cylinder are in the model)")
print("- H : _Mileage  0 (given Liter and Cylinder are in the model)")

mileage_f_type2 = anova_type2.loc['Mileage', 'F']
mileage_p_type2 = anova_type2.loc['Mileage', 'PR(>F)']

print(f"\nANOVA (typ=2) F-value for Mileage: {mileage_f_type2:.4f}, p-value:
    ↳ {mileage_p_type2:.4g}")

if mileage_p_type2 < 0.05:
    print(f"P-value = {mileage_p_type2:.6f} < 0.05, so we reject H")
    print("Conclusion: Adding Mileage significantly improves the model that
    ↳ already contains Liter and Cylinder")
```

```

else:
    print(f"P-value = {mileage_p_type2:.6f} 0.05, so we fail to reject H")
    print("Conclusion: Adding Mileage does not significantly improve the model,
    ↳that already contains Liter and Cylinder")

```

```

=====
PART (d): Partial (Type II) Sum of Squares
=====

```

Interpretation of 'Mileage' in Type II ANOVA:

- Reduced model: Price ~ Liter + Cylinder
- Full model: Price ~ Mileage + Liter + Cylinder
- Partial SS tests: Is 'Mileage' needed given that 'Liter' and 'Cylinder' are in the model?
- H : $\beta_{\text{Mileage}} = 0$ (given Liter and Cylinder are in the model)
- H : $\beta_{\text{Mileage}} \neq 0$ (given Liter and Cylinder are in the model)

ANOVA (typ=2) F-value for Mileage: 19.9048, p-value: 9.306e-06

P-value = 0.000009 < 0.05, so we reject H

Conclusion: Adding Mileage significantly improves the model that already contains Liter and Cylinder

```

[11]: print("="*50)
      print("PART (e): Manual ANOVA (typ=2) Test")
      print("="*50)

      null_model_e = smf.ols('Price ~ Liter + Cylinder',data=data).fit()
      full_model_e = smf.ols('Price ~ Mileage + Liter + Cylinder',data=data).fit()

      par_null = sm.stats.anova_lm(null_model_e, typ=2)
      par_full = sm.stats.anova_lm(full_model_e, typ=2)

      print("Null Model")
      print(par_null)
      print("\nFull Model")
      print(par_full)

      SSE_H0_e = par_null.loc['Residual', 'sum_sq']
      df_H0_e = par_null.loc['Residual', 'df']

      SSE_H1_e = par_full.loc['Residual', 'sum_sq']
      df_H1_e = par_full.loc['Residual', 'df']

      numerator_e = (SSE_H0_e - SSE_H1_e) / (df_H0_e - df_H1_e)
      denominator_e = (SSE_H1_e/df_H1_e)

      print(f"\nNumerator = ({SSE_H0_e} - {SSE_H1_e})/({df_H0_e} - {df_H1_e})")

```

```

print(f"Denominator = ({SSE_H1_e}/{df_H1_e})")

manual_F_stat_e = numerator_e/denominator_e

print(f"\nThe manually calculated F statistic is {manual_F_stat_e:.4f}")
print(f"\nANOVA (typ=2) F-value for Mileage from part (d): {mileage_f_type2:.4f}")
if np.isclose(manual_F_stat_e, mileage_f_type2):
    print("\nThis matches the F-statistic calculated from part (d)")
else:
    print("\nThis does not match the F-statistic calculated from part (d)")

```

```

=====
PART (e): Manual ANOVA (typ=2) Test
=====

```

Null Model

	sum_sq	df	F	PR(>F)
Liter	1.613541e+08	1.0	2.443669	0.118395
Cylinder	1.128963e+09	1.0	17.097866	0.000039
Residual	5.288960e+10	801.0	NaN	NaN

Full Model

	sum_sq	df	F	PR(>F)
Mileage	1.283997e+09	1.0	19.904763	0.000009
Liter	1.929760e+08	1.0	2.991552	0.084086
Cylinder	1.031948e+09	1.0	15.997457	0.000069
Residual	5.160560e+10	800.0	NaN	NaN

Numerator = (52889600780.65452 - 51605604120.22623)/(801.0 - 800.0)

Denominator = (51605604120.22623/800.0)

The manually calculated F statistic is 19.9048

ANOVA (typ=2) F-value for Mileage from part (d): 19.9048

This matches the F-statistic calculated from part (d)

```

[12]: print("\n" + "="*78)
      print("CODING QUESTION 5: KelleyBlueBookData Dataset Analysis")
      print("="*78)

```

```

=====
CODING QUESTION 5: KelleyBlueBookData Dataset Analysis
=====

```

```

[13]: print("="*50)
      print("PART (a): Model Fitting and Summary")

```

```

print("="*50)

# Fit the model
model_5 = smf.ols('Price ~ Mileage + Cylinder',data=data).fit()
print(model_5.summary())
print(f"\nThe R2 value from a model with Mileage and Cylinder as predictors is_
↪{model_5.rsquared}")
print(f"The Adjusted R2 value is {model_5.rsquared_adj}")

```

=====

PART (a): Model Fitting and Summary

=====

OLS Regression Results

```

=====
Dep. Variable:          Price    R-squared:                0.340
Model:                  OLS      Adj. R-squared:            0.338
Method:                 Least Squares    F-statistic:          206.2
Date:                  Sun, 28 Sep 2025    Prob (F-statistic):    5.95e-73
Time:                  21:59:43    Log-Likelihood:        -8369.2
No. Observations:      804    AIC:                  1.674e+04
Df Residuals:          801    BIC:                  1.676e+04
Df Model:               2
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	3145.7503	1325.934	2.372	0.018	543.034	5748.467
Mileage	-0.1524	0.035	-4.401	0.000	-0.220	-0.084
Cylinder	4027.6746	204.612	19.684	0.000	3626.036	4429.313

```

=====
Omnibus:                198.944    Durbin-Watson:          0.077
Prob(Omnibus):           0.000    Jarque-Bera (JB):       385.493
Skew:                    1.439    Prob(JB):               1.96e-84
Kurtosis:                4.797    Cond. No.:               1.01e+05
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.01e+05. This might indicate that there are strong multicollinearity or other numerical problems.

The R2 value from a model with Mileage and Cylinder as predictors is
0.33982070826263044

The Adjusted R2 value is 0.33817232051796786


```

[14]: print("="*50)
print("PART (b): Manual R2 and Adjusted R2 Calculations")
print("="*50)

SSE = model_5.ssr # or model_5.scale * model_5.df_resid
SST = model_5.centered_tss # Total sum of squares
n = model_5.nobs # Number of observations
p = model_5.df_model + 1 # Parameters (predictors + intercept)

print(f"\nSSE = {SSE}")
print(f"SST = {SST}")
print(f"n = {n}")
print(f"p = {p}")

r_squared = 1 - (SSE/SST)
r_squared_adj = 1 - ((SSE/(n-p))/(SST/(n-1)))

print(f"\nR2 = 1 - ({SSE}/{SST})")
print(f"R2 = {r_squared}")
print(f"\nAdjusted R2 = 1 - (({SSE}/({n}-{p}))/({SST}({n}-1)))")
print(f"Adjusted R2 = {r_squared_adj}")

if np.isclose(r_squared, model_5.rsquared) and np.isclose(r_squared_adj,
↪model_5.rsquared_adj):
    print("\nThese values match with the python outputs in (a) above")
else:
    print("\nThese values do not match with the python outputs in (a) above")

=====
PART (b): Manual R2 and Adjusted R2 Calculations
=====

SSE = 51798580167.8953
SST = 78461382864.00787
n = 804.0
p = 3.0

R2 = 1 - (51798580167.8953/78461382864.00787)
R2 = 0.33982070826263044

Adjusted R2 = 1 - ((51798580167.8953/(804.0-3.0))/(78461382864.00787(804.0-1)))
Adjusted R2 = 0.33817232051796786

These values match with the python outputs in (a) above

[15]: print("="*50)
print("PART (c): Model Fitting and Summary")

```

```

print("="*50)

# Fit the model
model_5c = smf.ols('Price ~ Mileage + Liter + Cylinder',data=data).fit()
print(model_5c.summary())
print(f"The R2 value from a model with Mileage, Liter and Cylinder as
↳predictors is {model_5c.rsquared}")
print(f"The Adjusted R2 value is {model_5c.rsquared_adj}")
print(f"The Adjusted R2 value increased by approximately {model_5c.rsquared_adj
↳- model_5.rsquared_adj}")

```

```

=====
PART (c): Model Fitting and Summary
=====

```

OLS Regression Results						
Dep. Variable:	Price	R-squared:	0.342			
Model:	OLS	Adj. R-squared:	0.340			
Method:	Least Squares	F-statistic:	138.8			
Date:	Sun, 28 Sep 2025	Prob (F-statistic):	2.18e-72			
Time:	21:59:43	Log-Likelihood:	-8367.7			
No. Observations:	804	AIC:	1.674e+04			
Df Residuals:	800	BIC:	1.676e+04			
Df Model:	3					
Covariance Type:	nonrobust					

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4707.6150	1602.866	2.937	0.003	1561.296	7853.934
Mileage	-0.1544	0.035	-4.461	0.000	-0.222	-0.086
Liter	1545.2522	893.411	1.730	0.084	-208.454	3298.958
Cylinder	2847.9345	712.040	4.000	0.000	1450.247	4245.622

Omnibus:	214.158	Durbin-Watson:	0.074
Prob(Omnibus):	0.000	Jarque-Bera (JB):	444.825
Skew:	1.499	Prob(JB):	2.56e-97
Kurtosis:	5.071	Cond. No.	1.37e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.37e+05. This might indicate that there are strong multicollinearity or other numerical problems.

The R2 value from a model with Mileage, Liter and Cylinder as predictors is 0.34228021178684886

The Adjusted R2 value is 0.33981376258104956

The Adjusted R2 value increased by approximately 0.0016414420630816995

R squared did increase when Liter was added as a predictor. However, this always happens as adding more predictors can never reduce the R squared value, the model will always fit at least as well (or better) when additional predictors are included. The adjusted R squared value also increased by approximately 0.0016414420630816995. This means that Model 2 is more preferable according to the adjusted R squared value, indicating that Liter as a predictor is worth adding. The adjusted R squared value penalizes for the addition of parameters. The fact that adding another predictor in Liter increased the adjusted R squared value means that there is a significant contribution from Liter as a predictor.

```
[16]: print("="*50)
      print("PART (d): Open question")
      print("="*50)
```

```
=====
PART (d): Open question
=====
```

From the type 2 ANOVA in question 4 part (c), the p-value was 0.084086 which is greater than 0.05, indicating that it is not significant. The adjusted R squared improvement is also not practically meaningful as the difference is about 0.16%. The t-stat from the OLS regression result is also very small (1.730), indicating that Liter's contribution is marginal given Mileage and Cylinder are already in the model. The lack of significance for Liter when Cylinder is in the model suggests multicollinearity between these predictors, as both measure engine size. It doesn't make sense to choose a more complex model for a 0.16% improvement in the adjusted R squared value, when the added predictor is not even significant. From a practical standpoint, both liter and cylinders measure the size of the engine so adding liter does not add meaningful pricing information. Therefore, choosing Model 1 makes the most sense in terms of the statistics and real-life meaning and applicability.